

Eric Liu

469-304-6182 | ericliu637@gmail.com | US Citizen
[ericliu637.github.io](https://github.com/ericliu637) | linkedin.com/in/eric-r-liu

EDUCATION:

University of Texas at Austin - Mechanical Engineer, Expected Graduation Date: May 2028

SKILLS:

Engineering: SolidWorks, Onshape, ANSYS (CFD, FEA, Thermal), KiCad

Software: Python, MATLAB, Bash, Ubuntu Linux, ESP32, Raspberry Pi, Arduino

Manufacturing and Prototyping: Soldering, 3D Printing

EXPERIENCE:

Research with University of Shanghai for Science and Technology (January 2026 - August 2026)

- Performed CFD modeling of gas flow inside an atomic layer deposition reaction chamber with ANSYS Fluent
- Optimized parameters such as injection times and chamber geometry, using Design of Experiments (DoE) to minimize precursor use and improve manufacturing throughput

Texas Design Build Fly (September 2024 - Present) | Systems Lead Engineer (August 2025 - May 2026)

- Directed a 15 member systems team through a full project lifecycle, overseeing drone payload mechanism integration, structural and reliability testing, and material selection, while increasing team retention by 60%
- Designed and executed testing program to determine the relationship between airspeed and drag of an aircraft-towed banner, analyzing data in Python to increase length by 133% and reduce ballast by 40%
 - Developed custom instrumentation using load cells, cameras, and pitot tubes
 - Performed root cause analysis of banner deployment failures, iterating folding geometry, banner rigging, and release mechanisms to improve deployment reliability
- Optimized loading time through 10 iterations of a quick attachment payload fairing that holds 30 lbs

IEEE Robotics and Automation Society (September 2024 - May 2025)

- Led a team to design and assemble a chess playing robot utilizing a CoreXY design, reducing footprint 50% while increasing movement speed

PROJECTS:

Robotics and Mechatronics

- Designed and built a pen plotter using geared DC motors and magnetic encoders (July 2026 - Present)
 - Implemented closed loop position control using a 1 kHz cascaded PID controller on an ESP32, with motion profiling and feedforward to improve movement accuracy
 - Validated performance and tuned control parameters by collecting position, velocity, error and motor data

Remote Controlled (RC) Vehicles

- Designed 9 custom RC aircraft for different mission profiles, such as short take off and landing (STOL), extremely low speed flight, or heavy payload capacity (June 2022 - Present)
 - Conducted CFD simulations in ANSYS Fluent of low Reynolds number airfoils and performed fluid structure interaction modeling with FEA to confirm structural integrity under load, verifying simulation accuracy with static structural testing
- Built an RC boat with a motorized camera module and neodymium magnet for underwater photography and object retrieval at a depth of 50 ft (May - December 2023)

Embedded Systems, Software, and AI

- Built a USB charge limiter using an ESP32 microcontroller, triggering a relay to stop charging when a user defined battery percentage is met to prolong battery life (July 2025)
- Developed an AI detection algorithm achieving 95% accuracy in benchmarks (February - August, 2025)
- Created and led a student software team developing a Chrome extension to improve usability of school grading system, growing adoption to 100+ active users (April, 2021 - August, 2023)